

## NOTES - Voice-Controlled Smart Lamp

### 1. Total build time:

Approximately 8–10 hours, including hardware setup, software installation, debugging, testing, Bluetooth microphone configuration, and implementation of additional features. Some additional time was spent troubleshooting and testing a USB headset microphone before switching to Bluetooth earbuds, which provided better speech recognition performance.

### 2. Microphone Used:

Bluetooth earbuds (built-in microphone) connected to the Raspberry Pi 5 via Bluetooth. The Bluetooth microphone provided better speech recognition performance than the available alternatives.

### 3. Backend Used:

Google Web Speech Recognition API

Reason:

The Google backend was selected because it provides good recognition accuracy, is easy to integrate using the SpeechRecognition Python library, and requires minimal configuration. An internet connection was available throughout testing, making it suitable for this project.

### 4. Measured End-to-End Latency

| Trial | Latency (s) |
|-------|-------------|
| 1     | 1.3         |
| 2     | 1.5         |
| 3     | 1.2         |
| 4     | 1.6         |
| 5     | 1.4         |
| 6     | 1.3         |
| 7     | 1.5         |
| 8     | 1.4         |
| 9     | 1.2         |
| 10    | 1.5         |

Average Latency: 1.39 seconds

## **5. Recognition accuracy: out of 20 attempted commands, how many succeeded?**

Out of 20 attempted voice commands, 18 commands were correctly recognized and executed.

Recognition accuracy: 90%

## **6. Commands consistently misrecognized:**

The following occasional recognition errors were observed:

- "Pink" → "Think"
- "Purple" → "People"

These errors mainly occurred when speaking softly or in the presence of background noise.

## **7. The most challenging parts for beginners are:**

- Finding and configuring a compatible microphone for Raspberry Pi. In this project, additional time was spent testing a USB headset microphone before switching to Bluetooth earbuds, which provided better performance.
- Installing the required Python libraries and configuring the audio system.
- Understanding how speech is converted into text and then translated into LED control commands.

## **Suggestions for Simplifying for a 9-Year-Old:**

- Provide a pre-configured Raspberry Pi with a working microphone already connected.
- Use color-coded wiring diagrams for the LED and push button.
- Use simple flowcharts or animations to explain how the microphone captures speech, the speech recognition converts it into text, and the Raspberry Pi controls the LED.

## **8. Extensions Implemented:**

The following additional features were implemented beyond the original project specification:

- Added additional LED colours to enhance the user experience and provide a wider range of voice-controlled lighting options. The newly added colours include Pink, Purple, Orange, White, and other custom colour variations.
- Developed a separate extension file named `extension_greeting.py`, which extends the functionality of the original project by supporting conversational voice commands such as:
  - Hello
  - Good Morning
  - Good Afternoon

- Good Evening
- Good Night
- Thank You

Each greeting command triggers a unique LED lighting pattern and displays an appropriate response message in the terminal, demonstrating that the system can perform interactive voice responses in addition to controlling LED colors.